

BUILD ARTIFACT 2: API CONTRACT APPENDIX (COMPLETE)

All 12 endpoints with full request/response schemas | FastAPI | Decimal Technologies

CONVENTIONS (apply to every endpoint)

Base path `/v1`. All bodies JSON. Every response carries `trace_id` (EVENT-RESP-004). All endpoints idempotent on `(trace_id, operation)`. Errors are structured: `{ error: {code, reason_enum, detail} }` — never a bare 500. Any endpoint that could take external action returns `requires_human_approval=true` at autonomy A1; only `/v1/send` executes externally, and only with a valid `human_approval_token`. Timestamps ISO8601 UTC.

1. POST `/v1/signals` — M1 Signal Ingestion (A2)

```
REQ { account_id:uuid|null, source:enum, signal_type:enum, priority:P1..P4,
      raw_context:str, source_url:str, detected_at:iso }
RES { signal_id:uuid, is_duplicate:bool, confidence:0..1, stored:bool,
      scoring_triggered:bool, trace_id:uuid }
ERR reason: INVALID_SOURCE | ENTITY_UNRESOLVED | MALFORMED_CONTEXT
```

2. POST `/v1/score` — M2 Scoring Engine (A1)

```
REQ { account_id:uuid, trigger_signal_id:uuid }
RES { dynamic_score:0..100, ics:0..100, static_fit_score:0..100,
      dimension_breakdown:{signal,regulatory,reachability,warmth:{raw,weight,contribution}},
      effective_opportunity:float, modifier_chain:{stage,budget,entrenchment,risk>window},
      decision_score:float, recommended_tier:enum, tier_changed:bool,
      requires_human_approval:true }
ERR reason: ACCOUNT_NOT_FOUND | ICS_BELOW_FLOOR (cannot promote, <40)
```

3. POST `/v1/enrich` — M3 Enrichment (A1)

```
REQ { account_id:uuid, depth:'standard'|'full' } // full if Fit>=80
RES { account_id:uuid, contacts:[{contact_id, name, title, seniority,
      email, email_verified:bool, linkedin_url,
      epis:{e,p,i,s}, source:'apollo'|'linkedin'}],
      buying_committee_complete:bool, // >=3 contacts (ENR-003)
      enrichment_id:uuid, legal_basis_code:str } // PDPL-001
ERR reason: NOT_TIER_1 | NO_VERIFIED_CONTACTS | PDPL_BASIS_MISSING
```

4. POST `/v1/generate-message` — M4 AI Chain (A1)

```
REQ { account_id:uuid, contact_id:uuid, signal_id:uuid, channel:'email'|'linkedin' }
RES { message_id:uuid, draft:str, channel:enum,
      reasoning_passport:{signal,interpretation,account_context,persona_analysis,
      pain_inference,strategy,qc_result}, // EXPLAIN-001
      evidence_strength:'FACT'|'INFERENCE'|'ASSUMPTION'|'SPECULATION',
      verifiability_score:0..100, qc_status:enum, regeneration_count:0..3,
      requires_human_approval:true }
ERR reason: CHAIN_TIMEOUT (>10min) | QC_FAILED_3X (escalate to human)
```

5. POST /v1/qc — M7 Quality Control (A1)

```
REQ { message_id:uuid, draft:str, account_id:uuid }
RES { qc_status:'PASSED'|"FAILED', rubric:{factual_accuracy:pass|fail,
      tone_match:1..5, personalization_depth:1..5, compliance:pass|fail,
      length_ok:bool}, average_score:float, // pass if >=3.5 AND all pass/fail OK
      flagged_claims:[{text, reason:'UNVERIFIED'|"BANNED_PHRASE', span:[i,j]}],
      hallucination_score:0..1 }
ERR reason: MESSAGE_NOT_FOUND
```

6. POST /v1/queue-message — M4 Ready-to-Fire Queue (A1)

```
REQ { message_id:uuid, signal_priority:P1..P4, signal_age_hours:int }
RES { queued:bool, queue_position:int, // ordered: priority then age (EVENT-RESP-003)
      earliest_send_window:iso, // next valid KSA window (QUEUE-ARCH-001)
      expires_at:iso, // 24h then regenerate (HUMAN-AUTO-002)
      awaiting:'human_review'|"send_window' }
ERR reason: QC_NOT_PASSED | DUPLICATE_IN_QUEUE
```

7. POST /v1/send — M4 Execution (A1, the only externally-irreversible call)

```
REQ { message_id:uuid, human_approval_token:uuid, reviewer_id:uuid }
RES { sent:true, sent_at:iso, channel:enum,
      accountability_record_id:uuid } // HUMANREVIEW-001 immutable
ERR 403 reason: NO_APPROVAL_TOKEN | OUTSIDE_SEND_WINDOW | CONTACT_EXHAUSTED |
               RAMADAN_BLACKOUT | CONTACT_OPTED_OUT | C_SUITE_NEEDS_L3_REVIEW
```

8. POST /v1/engagement — M8 Engagement Tracking (A2)

```
REQ { message_id:uuid, event_type:'open'|"click'|"reply'|"linkedin_accept',
      reply_text:str|null, occurred_at:iso }
RES { engagement_id:uuid, account_id:uuid, contact_id:uuid,
      intent_class:'POSITIVE'|"NEUTRAL'|"NEGATIVE'|"000'|"UNSUBSCRIBE', // ENG-002
      action:'HANDOFF'|"CONTINUE'|"NURTURE_COOLDOWN'|"RESCHEDULE'|"HARD_STOP',
      account_state_change:str|null, suspicious_engagement:bool } // bot detection
ERR reason: MESSAGE_NOT_FOUND
```

9. POST /v1/handoff — M9 Human Handoff (A1)

```
REQ { account_id:uuid, contact_id:uuid, trigger:'POSITIVE_REPLY'|"MEETING_REQUEST' }
RES { handoff_id:uuid, assigned_to:uuid,
      meeting_brief:{engagement_timeline, signal_history, contact_profile,
      talking_points, likely_objections, comparable_wins}, // INT-CRM-05
      slack_notified:bool, crm_task_id:uuid, ack_required_by:iso } // 24h ack
ERR reason: NO_OWNER_ASSIGNED | ALREADY_HANDED_OFF
```

10. POST /v1/sync-crm — M11 Distributed Reality Coordination (A2)

```
REQ { account_id:uuid, direction:'push' | 'pull' | 'both' }
RES { synced:bool, pushed_fields:[str], pulled_fields:[str],
      conflicts:[{field, pg_value, hubspot_value, resolution:'PG_WINS' |
                  'HUBSPOT_WINS' | 'LOCKED_72H'}], // CRM-003 source-of-truth split
      skipped:[{field, reason:'BELOW_MATERIALITY'}] } // CRM-ECON-01 (<5pt score)
ERR reason: HUBSPOT_AUTH_FAILED | RATE_LIMITED (backoff)
```

11. GET /v1/account/{id} — Account Read (A2, read-only)

```
RES { account_id:uuid, name, segment, state, tier,
      scores:{dynamic, ics, static_fit, effective_opportunity, decision_score},
      score_history:[{scored_at, dynamic, tier}],
      contacts:[...], active_opportunities:[...], active_sequences:[...],
      negative_decision_log:[{reason, rule_id, logged_at}] } // EXPLAIN-003
ERR 404 reason: ACCOUNT_NOT_FOUND
```

12. GET /v1/analytics/kpi — M10 Analytics (A2, read-only)

```
QUERY ?from=iso&to=iso&segment=str
RES { period:{from,to},
      north_star:{abm_sourced_pipeline_value}, // FAIL-WRONGMETRIC-001 hierarchy
      primary:{meetings_booked, positive_reply_rate},
      secondary:{total_reply_rate, signal_to_meeting_conversion},
      operational:{open_rate, click_rate, linkedin_accept_rate, qc_pass_rate},
      kpi_version:str } // INT-ANL-03 contracted + versioned
ERR reason: INVALID_DATE_RANGE
```

AUTONOMY SUMMARY ACROSS ENDPOINTS

A2 (act + audit, no external action): /signals, /engagement, /sync-crm, /account, /analytics. A1 (recommend, human decides): /score, /enrich, /generate-message, /qc, /queue-message, /handoff. A1 with hard external gate: /send (requires human_approval_token; C-suite never auto-sends per AUTO-CEILING-001). This mapping is the wire-level enforcement of the Autonomy Ladder.